
Simulating Realistic Player Conversations with Controllable Properties under Privacy Constraints

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Abstract

Generating human-aligned synthetic conversational data with large language models (LLMs) is attractive when privacy constraints limit access to real text, but practical recipes for privacy-aware prompting and their downstream utility remain unclear. We study minimal-disclosure conditioning—prompting with aggregate statistics (e.g., mean/SD of utterance length, most frequent words per label) and token co-occurrence cues—across three settings (Reddit GoEmotions sentiment/emotion, DOTA2 toxicity, and Call of Duty prosociality in-game chats). We compare these to naïve class-conditional prompting without auxiliary distributional information, evaluate distributional alignment to real data via MMD and Wasserstein distance, and measure downstream utility by training a low-capacity classifier on the synthetic corpora. On DOTA 2, co-occurrence cues provide modest gains over naïve prompting (AUC 0.83 vs. 0.76; precision 0.09 vs. 0.08; recall 0.83 vs. 0.78), while summary-statistic prompts yield greater improvements (AUC 0.89; precision 0.13; recall 0.88); however, across datasets we observe no consistent improvements in distributional similarity. Importantly, classifiers trained on synthetic data outperform models trained on small real subsets (AUC 0.81–0.82 vs. 0.45–0.75 with 10–50 real samples) and match or exceed training on 100–200 real examples (0.77–0.79), highlighting a practical data-efficiency benefit under privacy constraints. We discuss limitations and non-transferability across domains, and outline opportunities for human-centered deployment of synthetic data in gaming applications.

1 Introduction

In contexts with human conversational datasets, there is a clear and present need for labeled examples of human data. In most cases, there is not enough human labeled data that can be used for a variety of use cases. To supplement this lack of real human-labeled examples, the goal of this study is to use LLM’s to generate synthetic examples that mimic the real distribution in order to supplement training sets. The specific use case will be to study LLM’s ability to simulate human conversations in social media and in gaming. Specifically, video game chats and Reddit comments were analyzed and synthetic examples for those datasets were generated. However, it is not possible to just give real

examples to the LLM and ask it to generate similar samples is that there are a multitude of privacy concerns that govern the use of this real world data. In order to still be able to generate high quality synthetic data, the goal of this study is to see if privacy preserving methods can be used to convey enough information about the distribution to be able to generate high quality synthetic data that is in distribution to real world data.

1.1 Prior Work

Earlier methods to generate synthetic text data relied on Generative Adversarial Networks (GAN) or Variational Autoencoders, but these techniques do not provide as much control over the generations as LLMs, along with another host of issues. Specifically, when looking at GAN's, they struggle with mode collapse, non-convergence and instability [1]. Variational Autoencoders also present issues with no variance in the generations which will present problems in training weak classifiers [2]. These issues lead toward utilizing LLM's to solve this problem. Yet LLMs themselves also suffer from challenges, in a zero-shot setup they may not model the desired real-world distribution well, especial if the data is unusual or underrepresented in the training set (e.g., private gaming chat). When relying on examples (few-shot), the LLMs tend to stick to only shallow-variations of provided examples [3, 4], limiting the generalization of the synthetic variations to unseen data [5].

1.2 Problem Area

This study looks to explore the issues surrounding the generation of synthetic data, specifically text data, under privacy constraints. This text data needs to be as close to the real distribution that it is based off of, while still allowing for controllable properties to ensure that the introduction of synthetic data doesn't cause overfitting to the provided examples in the model generations. One way to generate this realistic synthetic data is to communicate information about the real data distribution to the generator. There is a balance that must be found between communicating pertinent information to the generator while still ensuring that the privacy of the users' information is preserved. That leads to the idea of looking to summarize the real distribution in the best way possible to ensure similar examples that are in-distribution, while ensuring that user's privacy is preserved.

1.3 Findings

The main findings in this study revolve around the performance of various privacy preserving techniques as well as the difference in performance of different types of datasets. In regards to the privacy preserving techniques, the summary statistics seem to perform the best across all the datasets. This method also outperforms the LLama baseline by 0.35 and the Plug-and-Play baseline by 0.13 as evident by Table 4. Also, it can be seen that the Token Co-Occurrence method outperformed the Zero-Shot prompting by 0.13 for the DOTA 2 dataset (evident by Table 2, showing that communication the distribution of the dataset to the LLM lead to improvement of the weak classifier. In comparison to the other datasets in the study, the performance of the DOTA 2 dataset slightly outperformed the others for the Token Co-Occurrence and Summary Statistics, but the COD outperformed the Reddit by 0.01 and the DOTA 2 by 0.04 according to Table 2

1.4 Contributions

This is one of the first works that combines privacy preserving data summarization in conjunction with LLM generating in order to generate synthetic data. This study also explored the use of an LLM to generate synthetic data based off of proprietary data not found in the LLM's training set (COD data).

Methods

1.5 Datasets

The main datasets that were used in this study were the Reddit GoEmotions, the DOTA 2 Matches, and the proprietary COD chat dataset. The Reddit and DOTA 2 are large public datasets, while the COD is a proprietary dataset provided by Activision. The Reddit dataset was labeled for positive and

negative emotions, as the dataset originally consisted of Reddit comments labeled with one out of 28 emotions. The DOTA 2 dataset consists of chats labeled either toxic or non-toxic as defined by video game moderators. As for the COD dataset, it consists of chats labeled for prosocial or non-prosocial, where pro-sociality is defined by the following behaviors: helping and defending, assisting or acting in someone else’s interest, volunteering, donating, and comforting, cooperation and sharing, and expressing empathy and emotional awareness [6]. There were 2000 examples sampled from each dataset with 1000 of those being training examples and 1000 for testing. Out of all the datasets, the DOTA 2 contains a large class imbalance with about 4% of the labels being toxic while 96% being non-toxic.

Dataset	Messages	Negative Label Counts	Positive Label Counts	Avg. Word Count
DOTA 2	3, 460	3321 examples	139 examples	2.61 words
Reddit	41, 701	25711 examples	15990 examples	13.09 words
COD	500	320 examples	180 examples	30.62 words

Table 1: Overview of the datasets that were used in this study. The positive label for DOTA 2 refers to toxic chats, for Reddit it refers to negative emotions, and for COD the positive label is for prosocial behavior.

1.6 Privacy Preserving Summarization Methods

The privacy preserving methods for this study were selected due to their application in practical scenarios and their explainability. This allows for quick implementation which might be necessary to solve business problems in industry.

Summary Statistics The summary statistics method gave two important pieces of information to the LLM, the mean and standard deviation of the lengths of the messages as well as the most frequent words for each label. The prompt for this technique included information on what kind of language would be found in the specific dataset as well as a definition on each label type.

The actual pipeline to generate the synthetic examples first consisted of taking the training corpus, parsing it to get frequency counts for all the words found in each label class. From there, the top 100 for each label were selected and given to the LLM. Along with this, the mean and standard deviation of the message lengths were also passed to the LLM. These pieces of information were provided to the prompt (Table 6) which then generated the examples.

Word co-occurrence The Word Co-Occurrence method provides information on what words are near each other to the the LLM to base its generations off of. Specifically, the co-occurrences are in respect to words within the same message. The reason it was selected as one of the privacy preserving techniques to summarize data was because its an explainable method and it is cheap to implement in practice. Also, it was chosen because it preserves the statistical structure of the original text without exposing sensitive raw content, making it both safe and useful for synthetic data generation.

The actual pipeline to generate the synthetic examples first consisted of taking the training corpus, parsing it to take all the co-occurring words and their frequencies and placing them into a dictionary. From there, the co-occurrences that contained stop words were removed and out of the remaining pairs, the top 100 were taken and they were provided to the prompt (Table 5) which then generated the examples.

1.7 Baselines/Comparisons

The main baselines for this project consisted of using a smaller model to generate synthetic data, not providing any information to the LLM and asking it to simply generate what it thinks to be in-distribution data, and using Plug-and-Play Language Models [7], which are a popular class of models used to solve this problem. Another baseline method, back-translating, was also utilized as an upper bound on the performance of the transformer based linear probe.

Zero-shot: In this method, the LLM was given direction as to the type of data that it was supposed to generate as well as descriptions of the labels. There was no other summary statistics given in order to determine how well the LLM would perform without any direction.

Plug-and-Play: In this method, there is a base level LLM that is provided plugged-in pieces that steer the output into the required direction. Specifically, there is a Qwen2.5B-Instruct model that is given specific plug-ins relating to the types of labels as well as the context as to what type of content the LLM needs to generate [8]. Along with this, there are specific instructions to remove emojis, hashtags and phrases that are not supposed to be in the generation.

Upper bound - Back-Translate: In this method, examples were taken and translated into French and from there translated back into English. This technique was to serve as an upper bound for the synthetic data performance for the weak classifier. The reason for this is that the back-translating would preserve the original meaning of the message and so therefore wouldn't really present "new" examples of chat content, just old examples that were slightly augmented.

1.8 Study Design

This study consisted of a segmented pipeline that would take the statistics gathered from the real data and ultimately lead to synthetic examples that were generated by the LLM. To start, the original datasets were split into a training and testing sample, with about 1000 examples in each for the DOTA 2 and the Reddit and 250 examples in each for the COD. The reason for selecting precisely 1000 examples was due to the fact that taking any less would not allow for enough exposure to minority class examples as most real world data sets have a large amount of class-imbalance. However, taking too many examples would not lead to a fair comparison as there is a limited amount of labeled data for the COD dataset (only a total of 500 examples) and so the other two datasets were capped out at a total of 2000 examples to keep the comparisons as close as possible. After selecting the examples, the appropriate summary statistics were gathered from the training sample and passed into the prompt to be given to the LLM. With this, the LLM was tasked to generate 3000 synthetic examples for the COD and DOTA 2 and 750 synthetic examples for COD, in batches of 10 examples. From there, these examples were passed to the downstream task, where they were to be used to train a SVM classifier. The examples were split into three groups, with each group going through the training pipeline separately. This classifier was then evaluated on the real testing sample gathered earlier. This was done for each group to get 3 results, from which an average was taken along with a standard deviation. In order to do this process, the messages were converted to embeddings using the all-mpnet-base-v2 transformer [9]. On the testing set, the ROC-AUC was measured along with the precision and recall of the model.

After collecting these measures from the original dataset, the next step was to perform the same training process, but this time inject 10% of the real examples into the training. This was done by taking 100 examples from the training sample with 900 of the synthetic and using that 1000 message sample to train the weak classifier. Again, this was repeated for the three groups of synthetic data and the average and standard deviation of the results was computed.

Metrics The main metrics measured in this study consisted of two types, the distributional measures between the synthetic and real data and the metrics for the downstream task, i.e the performance of the weak classifier. For the distributional measures, the Maximum Mean Discrepancy (MMD) and Wasserstein Distance were considered. The MMD gives a measure of how far the Mean of the two distributions are from each other, while the Wasserstein computes the cost of transforming one distribution into another.

Along with the two distributional measures, the performance of the weak classifier was measured through the ROC-AUC score, Precision, and Recall of the SVM classifier on the real testing sample were measured. This looked to give a reading on how well the classifier was able to perform in correctly labeling messages for the given task.

2 Results

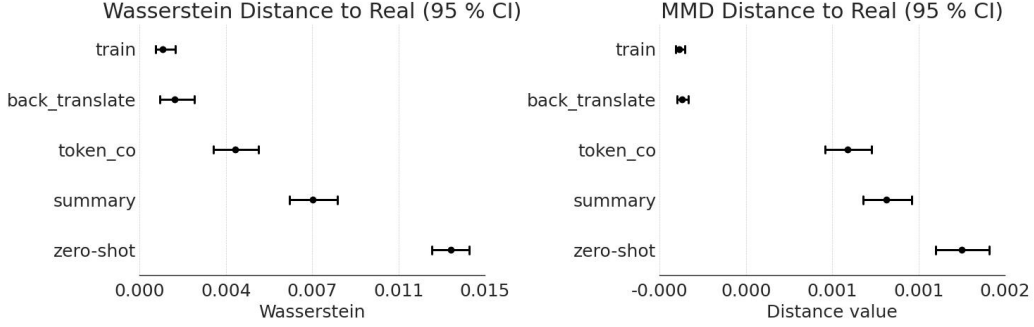


Figure 1: This is a plot of the distributional distances comparing the synthetic data to the real test data for the DOTA 2 data. The train row corresponds to the original training data, the back-test to the upper bound experiment, and the three remaining rows to the summarizing techniques.

Privacy Preserving Technique	Reddit	Improv.	DOTA 2	Improv.	COD	Improv.
Zero-shot	0.79 ± 0.01	–	0.76 ± 0.05	–	0.80 ± 0.03	–
Summary Stats	0.81 ± 0.00	$\uparrow 2.53\%$	0.89 ± 0.01	$\uparrow 17.1\%$	0.83 ± 0.02	$\uparrow 3.75\%$
Word Co-Occurrence	0.81 ± 0.00	$\uparrow 2.53\%$	0.83 ± 0.01	$\uparrow 9.21\%$	0.78 ± 0.02	$\downarrow 2.5\%$
Zero-shot + 10%	0.78 ± 0.02	–	0.73 ± 0.05	–	0.84 ± 0.02	–
Summary Stats + 10%	0.81 ± 0.00	$\uparrow 3.85\%$	0.86 ± 0.02	$\uparrow 21.92\%$	0.86 ± 0.03	$\uparrow 2.38\%$
Word Co-Occurrence + 10%	0.81 ± 0.01	$\uparrow 3.85\%$	0.78 ± 0.05	$\uparrow 6.85\%$	0.82 ± 0.05	$\downarrow 2.38\%$

Table 2: Overview of key results for the different privacy preserving techniques

Subset	Reddit	Improv.	DOTA	Improv.	COD	Improv.
Full Synthetic	0.81 ± 0.00	–	0.89 ± 0.01	–	0.83 ± 0.02	–
200	0.79 ± 0.01	$\downarrow 2.47\%$	0.87 ± 0.06	$\downarrow 2.25\%$	0.88 ± 0.00	$\uparrow 6.02\%$
100	0.77 ± 0.03	$\downarrow 4.94\%$	0.87 ± 0.08	$\downarrow 2.25\%$	0.86 ± 0.02	$\uparrow 3.61\%$
50	0.72 ± 0.03	$\downarrow 11.11\%$	0.86 ± 0.08	$\downarrow 3.37\%$	0.82 ± 0.05	$\downarrow 1.20\%$
10	0.45 ± 0.18	$\downarrow 44.44\%$	0.29 ± 0.01	$\downarrow 67.42\%$	0.47 ± 0.28	$\downarrow 42.37\%$

Table 3: Key results for results of classifier trained on full synthetic vs subsets of real data.

Baseline	AUC
Llama Baseline	0.54 ± 0.05
Plug and Play Baseline	0.76 ± 0.01

Table 4: Summary of the main baselines to compare against for the project. These results are for the DOTA 2 dataset

2.1 Distribution Similarity

Looking at the distributional results in Figure 1, we can see that the data summarizing methods seem to provide data that is closer to the real distribution than the zero-shot method. Along with this, it can be seen that the back translating and real training set outperforms the summarizing methods.

This is expected as these methods are not synthetic and all, so they provide a sort of measuring stick to compare against. Regarding the difference between the word co-occurrences and summary statistics, the plots indicate that the word co-occurrence method generates data that is closer to the real distribution than the summary statistics method.

2.2 Downstream Classification Task

For the downstream classification task, the key results that can be interpreted from Table 2 are that the best synthetic data that was generated came from the Summary Statistics technique applied on the DOTA 2 data, which lead to an AUC of 0.89. Another result from this study is that both techniques outperformed the zero-shot method, with the summary stats outperforming by 7.79% averaged across all 3 datasets and the word co-occurrences outperforming by 3.08% average across all 3 datasets. An interesting thing to note is that the word co-occurrences failed to outperform the zero-shot for the COD dataset, underperforming by 2.5%.

With the addition of 10% real data, there were a multitude of results to consider. Namely, there seemed to be a decrease in performance for the Reddit and DOTA 2 datasets, with the drop being as low as 6.02% for the word co-occurrence DOTA 2 result. However, for the COD there is a performance increase, with it being as large as 5.13% for the word co-occurrence result. The dataset that had the least effect of the increase was the Reddit, with performance decreasing about 0.42%.

For the subset table, the main results that can be seen is that for the most part, the Synthetic data outperforms the smaller subsets of real data. This can be seen, in the case of the Reddit and DOTA, where the full synthetic outperforms 200 real examples by 2.47% and 2.25% respectively. However, for the COD data, the Full Synthetic underperforms the 200 and 100 real examples by 6.02% and 3.61% respectively.

3 Discussion

3.1 Different Behavior in Different Datasets

The main difference in behavior that can be observed in the results can be seen in the results of the COD and the other two datasets. The exact difference in behavior that can be seen is in the difference in performance between the full synthetic and the full synthetic + 10% real. As seen in Table 2, the performance of the COD improves with the introduction of real data, whereas in the other datasets it decreases. This is potentially due to the fact that the Reddit and DOTA 2 are public datasets and therefore have been consumed by the LLM previously when it was being trained. The COD on the other hand is a private dataset that the LLM has not been exposed to. This indicates that the LLM potentially wasn't able to fully capture the essence of the COD data and therefore the real data was able to supplement cases that the LLM did not supply.

3.2 Impact of mixing in real data for Reddit and DOTA 2

The impact of mixing in real data provided a interesting result for the Reddit and DOTA 2 datasets. Specifically, the Reddit and DOTA 2 datasets underperformed with the introduction of the real data. This can potentially be explained by similar logic above, this time with the LLM being able to capture more of the public datasets. The introduction of the real data might actually cause the model to loose generality to some less popular cases and thus lose performance.

3.3 Length of Messages to Performance

When looking at the length of the messages, to their performance, it can be seen that both summarizing techniques had a larger positive impact on performance for shorter datasets. This can be see as the DOTA 2, which is has smallest number of words on average according to Table 8, saw the largest improvement to performance with jumps of 17.1% and 9.21% for the summary stats and word co-occurrences respectively. This might be due to the fact that the LLM is more accustomed to generating longer pieces of text with better grammar, and that communicating information about more unfamiliar text distributions to an LLM will lead to larger performance increases than for text distributions that are more similar to LLM generations.

3.4 Future Work

Future work for this study consists exploring a more complex summarizing method for distributions. One example would be to communicate SHAP-informed keywords to the LLM to base its generations off of. Along with this, another idea is to build off of the summary statistics by doing per-instance sampling of a real distribution of text and then passing that sample to the LLM to generate its text off of.

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A Technical Appendices and Supplementary Material

Table 5: **Prompt for Token Co-Occurrence DOTA 2 Chat Generations.**

Feedback Generation Prompt
Prompt. You are a data generator tasked with creating realistic DOTA 2 chat messages. These chat messages should be labeled according to their sentiment: toxic or non-toxic. Base the style on typical video game chat messages — include informal internet language, typos, and abbreviations. You will be given statistics about the distribution, including average chat length, standard deviation, and most common words associated with each label and their frequency. Generate exactly 10 realistic DOTA 2 chat messages, one per line. Each line should follow this format: the chat message in double quotes, followed by a space and then the label (1 for toxic, 0 for non-toxic). No extra formatting — just plain text output, one line per comment. Here is the format: <pre>gg dawg 0 Your trash 1</pre> Inputs • <i>Co-Occurrences</i> (list of strings): List of messages that give the most co-occurring words that appear with each other. Output • (str) 10 DOTA 2 chat messages containing a message and the corresponding label.

Table 6: **Prompt for Summary Stats DOTA 2 Chat Generations.**

Feedback Generation Prompt
Prompt. You are a data generator tasked with creating realistic DOTA 2 chat messages. These chat messages should be labeled according to their sentiment: toxic or non-toxic. Base the style on typical video game chat messages — include informal internet language, typos, and abbreviations. You will be given statistics about the distribution, including average chat length, standard deviation, and most common words associated with each label and their frequency. Generate exactly 10 realistic DOTA 2 chat messages, one per line. Each line should follow this format: the chat message in double quotes, followed by a space and then the label (1 for toxic, 0 for non-toxic). No extra formatting — just plain text output, one line per comment. Here is the format: <pre>gg dawg 0 Your trash 1</pre> Inputs • <i>Avg-len</i> (float): The average length of a message. • <i>Std-len</i> (float): The standard deviation of the length of a message. • <i>Common Words</i> (list of strings): A list of the most frequent words for each label class Output • (str) 10 DOTA 2 chat messages containing a message and the corresponding label.

Privacy Preserving Technique	Reddit	Gain	DOTA 2	Gain	COD	Gain
Zero-shot	0.79 $\pm$ 0.01	—	0.76 $\pm$ 0.05	—	0.80 $\pm$ 0.03	—
Summary Stats	0.81 $\pm$ 0.00	$\uparrow$ 2.3%	0.89$\pm$0.01	$\uparrow$ 17.1%	0.83$\pm$0.02	$\uparrow$ 3.8%
Word Co-Occurrence	0.81 $\pm$ 0.00	$\uparrow$ 2.5%	0.83 $\pm$ 0.01	$\uparrow$ 9.2%	0.78 $\pm$ 0.02	$\downarrow$ 2.5%
Zero-shot + 10%	0.78 $\pm$ 0.02	—	0.73 $\pm$ 0.05	—	0.84 $\pm$ 0.02	—
Summary Stats + 10%	0.81 $\pm$ 0.00	$\uparrow$ 3.9%	0.86$\pm$0.02	$\uparrow$ 21.9%	0.86$\pm$0.03	$\uparrow$ 2.4%
Word Co-Occu. + 10%	0.81 $\pm$ 0.01	$\uparrow$ 3.9%	0.78 $\pm$ 0.05	$\uparrow$ 6.9%	0.82 $\pm$ 0.05	$\downarrow$ 2.4%

Table 7: Overview of key results for the different privacy preserving techniques

Dataset	Messages	# Neg. Label	# Pos. Label	Avg. Word Count
DOTA 2	3,460	3321	139	2.61
Reddit	41,701	25,711	15,990	13.09
COD	500	320	180	30.62

Table 8: Overview of the datasets that were used in this study. The positive label for DOTA 2 refers to toxic chats, for Reddit it refers to negative emotions, and for COD the positive label is for prosocial behavior.